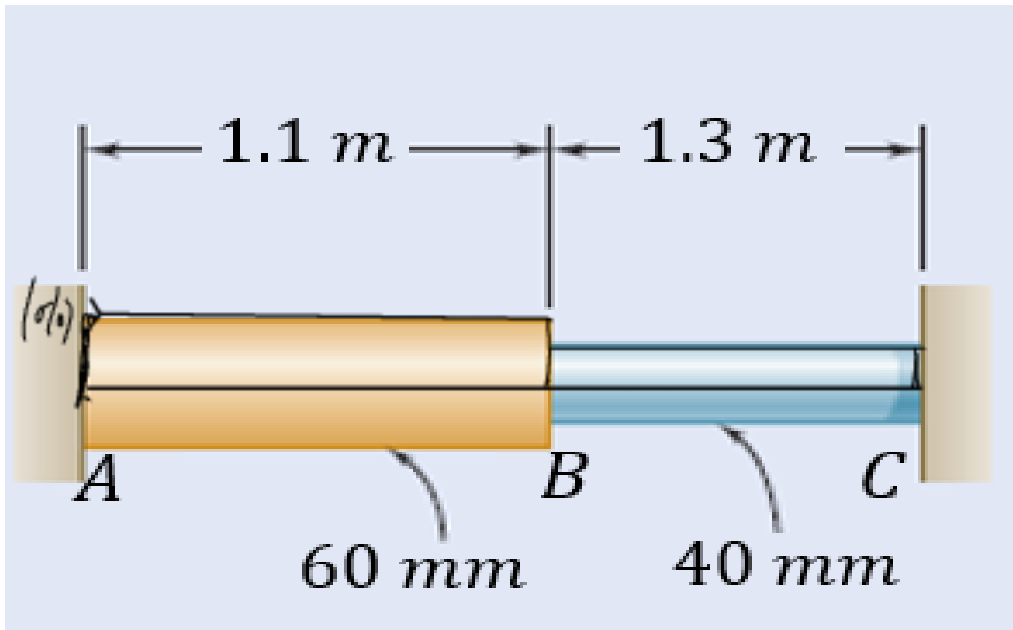


1st Practical Work: Axial and thermal loads

A. Thermal stresses in a compound rod made of two different materials

A rod consisting of two cylindrical portions AB and BC is restrained at both ends. Portion AB is made of brass ($E_b = 105\text{ GPa}, \alpha_b = 20.9 \times 10^{-6}/^{\circ}\text{C}$) and portion BC is made of aluminum ($E_a = 72\text{ GPa}, \alpha_a = 23.9 \times 10^{-6}/^{\circ}\text{C}$). Knowing that the rod is initially unstressed, the objective is to determine the stresses induced in AB and BC , as well as the displacement of the point B , when there is a temperature rise of 42°C . The length and diameter of each portion are shown in the figure.



1. Analytical solution (calculations by hand):
- a. Determine the normal stresses in the section AB & BC , σ_{AB} and σ_{BC}

b. Determine the displacement of the point B ($\delta_B = \delta_T^{AB} - \delta_R^{AB}$)

i. Calculate the total elongation (change of length) δ_T due to temperature that the rod would have had if it wasn't restricted. ($\delta_T = \delta_T^{AB} + \delta_T^{BC}$)

ii. Determine (as a function of R) the change of length δ_R of the rod due to the reaction force R from the supports ($\delta_R = \delta_R^{AB} + \delta_R^{BC}$)

iii. Knowing that $\delta_T = \delta_R$, find the value of R and deduce the resulting normal stresses σ_{AB} and σ_{BC} in the two portions of the rod.
2. Numerical solution (using the software, follow the provided guide):
- Using Abaqus (Educational Edition), launch a "Abaqus CAE":
- a. Determine the normal stresses σ_{AB} and σ_{BC} in the axial direction in the middle of the portions,

b. and the displacement (deformation) of the point B along the axial direction
3. Compare the analytical and numerical solutions.

	1. Analytical solution:	2. Numerical solution:
Normal stress in AB section	$\sigma_x = \dots\dots\dots$ MPa	$S_{11} = \dots\dots\dots$ MPa
Normal stress in BC section	$\sigma_x = \dots\dots\dots$ MPa	$S_{11} = \dots\dots\dots$ MPa
Displacement at point B:	$\delta_B = \dots\dots\dots$ mm	$U_1 = \dots\dots$ mm

1. Analytical Solution

a). Determine the normal stresses in the section AB & B

i). Calculate the total elongation δ_T

$$\begin{aligned}\delta_T &= \delta_T^{AB} + \delta_T^{BC} \\ &= \alpha_b L_b \Delta T + \alpha_a L_a \Delta T \\ &= (20.9 \times 10^{-6} \times 1100 \times 42) + (23.9 \times 10^{-6} \times 1300 \times 42) \\ &= 0.9664 \text{ mm} + 1.3003 \\ \delta_T &= 2.2668 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Since } \alpha_b &= 20.9 \times 10^{-6} / ^\circ\text{C} \\ L_b &= 1.1 \text{ m} = 1100 \text{ mm} \\ \alpha_a &= 23.9 \times 10^{-6} / ^\circ\text{C} \\ L_a &= 1.3 \text{ m} = 1300 \text{ mm}\end{aligned}$$

ii) Determine change in length due to reaction force R

$$\begin{aligned}S_R &= S_R^{AB} + S_R^{BC} \\ &= \frac{R L_b}{E_b A_b} + \frac{R L_a}{E_a A_a} \\ &= R \left(\frac{1100}{105 \times 10^3 \times 2827.43} + \frac{1300}{72 \times 10^3 \times 1256.64} \right) \\ S_R &= R \times 18.084 \times 10^{-6} \text{ mm/N}\end{aligned}$$

$$\begin{aligned}\text{Since } E_b &= 105 \text{ GPa} = 105 \times 10^3 \text{ MPa} \\ d_b &= 60 \text{ mm} \\ E_a &= 72 \text{ GPa} = 72 \times 10^3 \text{ MPa} \\ d_a &= 40 \text{ mm}\end{aligned}$$

$$\begin{aligned}\Rightarrow A_b &= \frac{\pi}{4} d_b^2 = \frac{\pi}{4} (60)^2 = 2827.43 \text{ mm}^2 \\ A_a &= \frac{\pi}{4} d_a^2 = \frac{\pi}{4} (40)^2 = 1256.64 \text{ mm}^2\end{aligned}$$

iii) Find Value R and deduce Stress

$$\begin{aligned}\Rightarrow 2.2668 &= R \times 18.084 \times 10^{-6} \\ \Rightarrow R &= \frac{2.2668}{18.084 \times 10^{-6}} = 125345 \text{ N} = 125.35 \text{ kN}\end{aligned}$$

$$\begin{aligned}\Rightarrow \sigma_{AB} &= \frac{-F}{A_b} = \frac{-125345}{2827.43} = -44.39 \text{ MPa} \\ \sigma_{BC} &= -\frac{F}{A_a} = -\frac{125345}{1256.64} = -99.74 \text{ MPa}\end{aligned}$$

B). Find displacement of point B

$$\begin{aligned}S_B &= S_T^{AB} - S_R^{AB} = (\alpha_b L_a \Delta T) - \frac{R L_b}{E_b A_b} \\ &= 0.96642 - (3.705 \times 10^{-6} \times 125345) \\ &= 0.502 \text{ mm}\end{aligned}$$

$$\underline{S_B = 0.502 \text{ mm}}$$

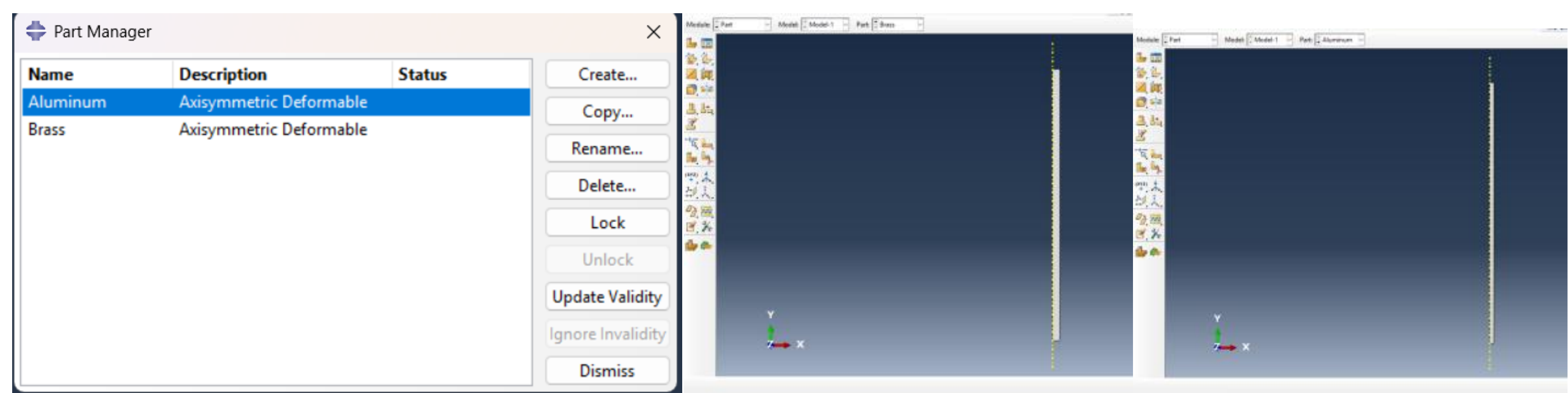
2. Numerical solution (using the software, follow the provided guide):

Using Abaqus (Educational Edition), launch a “Abaqus CAE”:

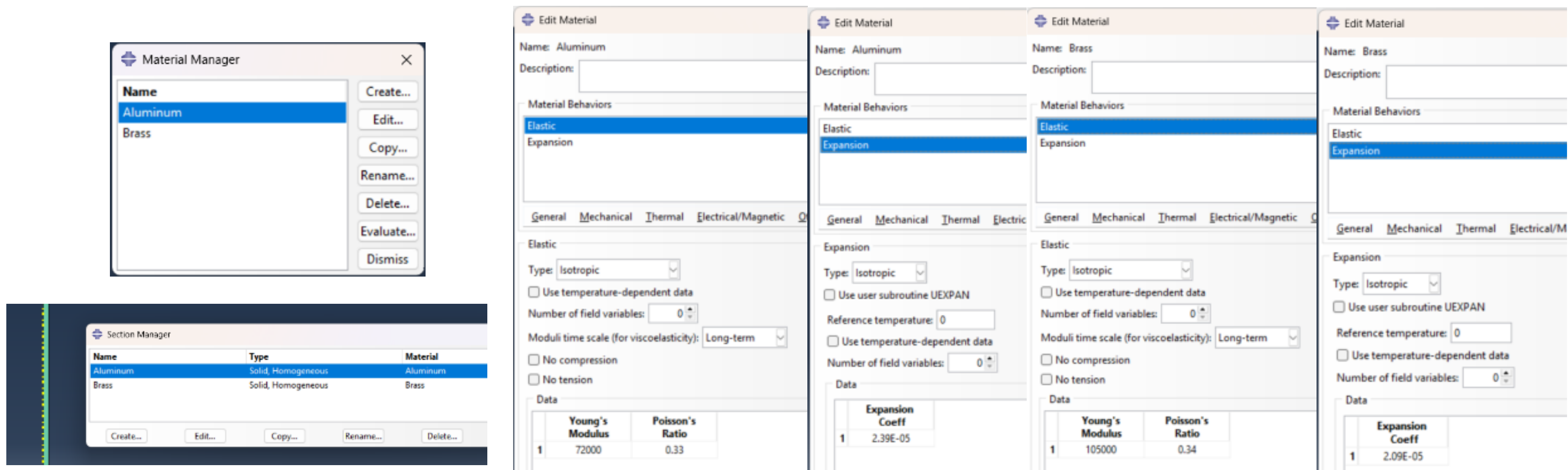
- a. Determine the normal stresses σ_{AB} and σ_{BC} in the axial direction in the middle of the portions,
- b. and the displacement (deformation) of the point B along the axial direction

Numerical Solution

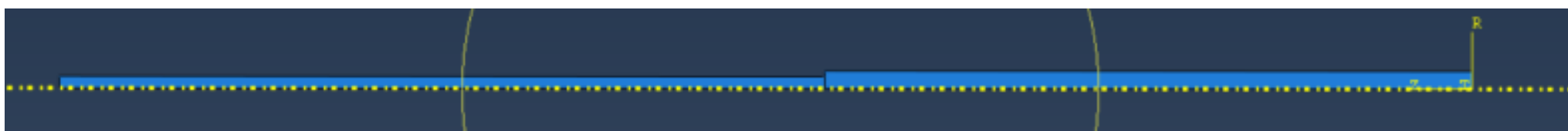
- a. Determine normal stress in the axial direction in the middle of both portions.
 - First define the parts:



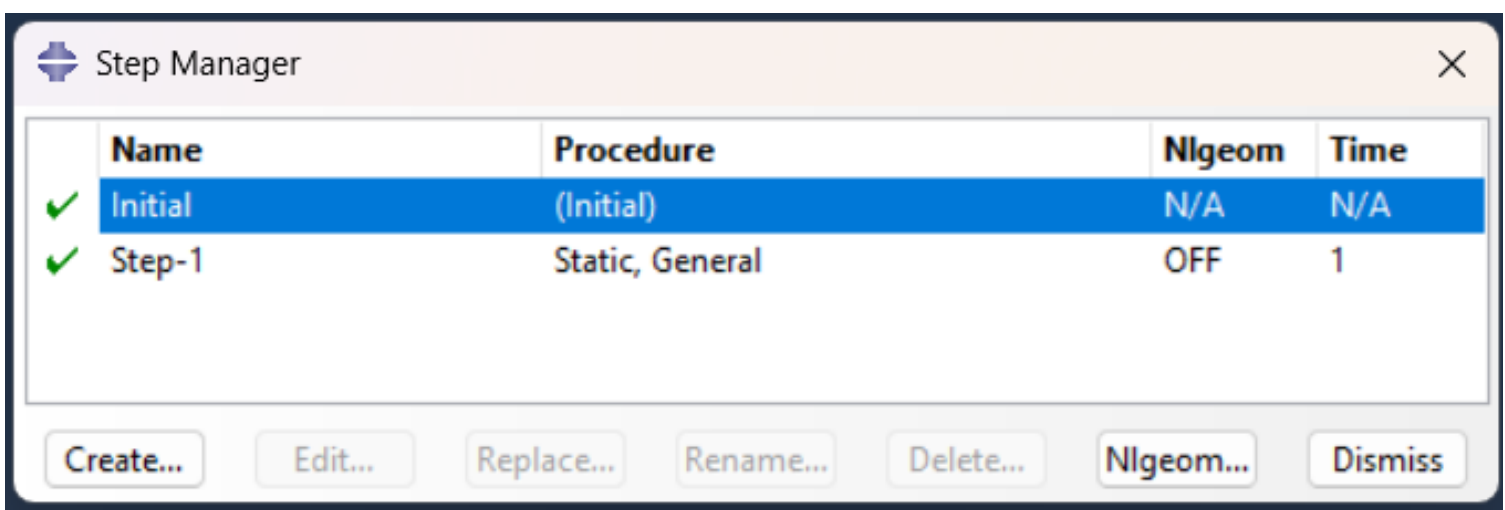
- Property



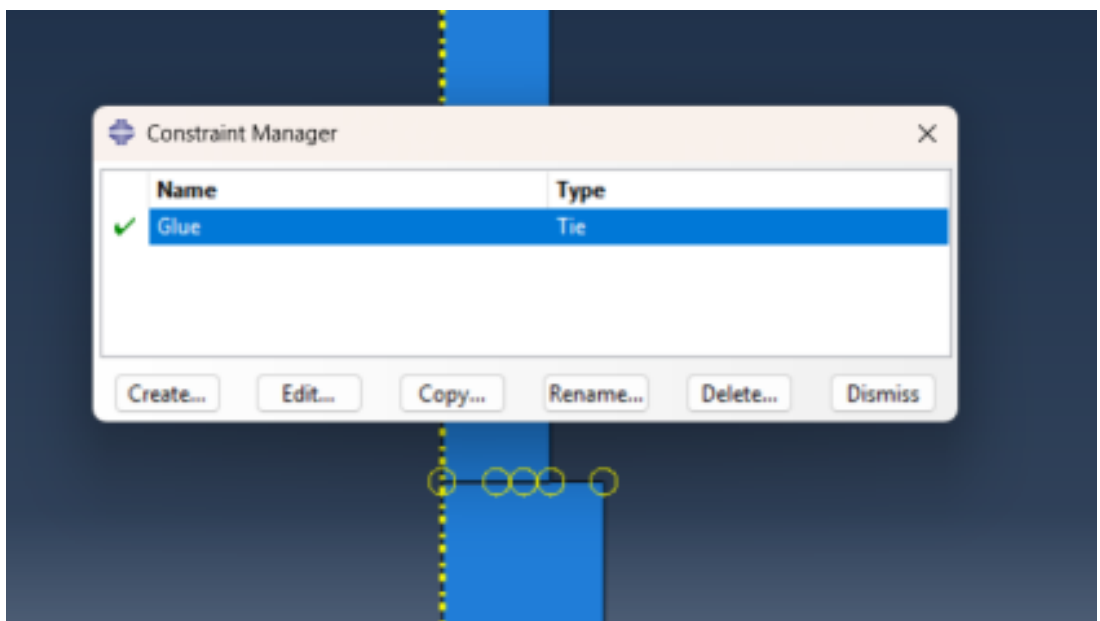
- Assemble



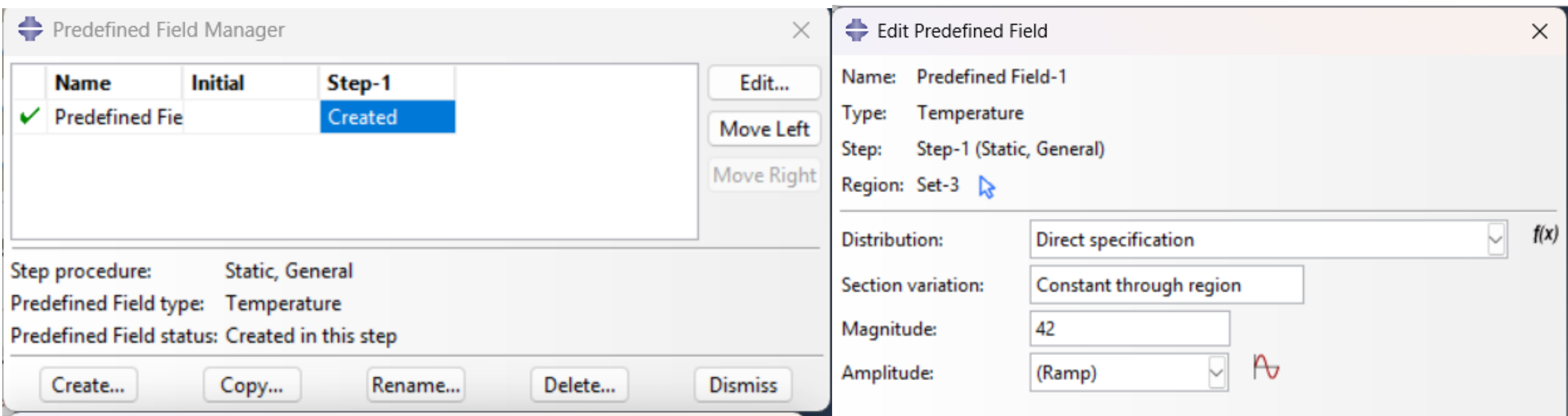
- Step



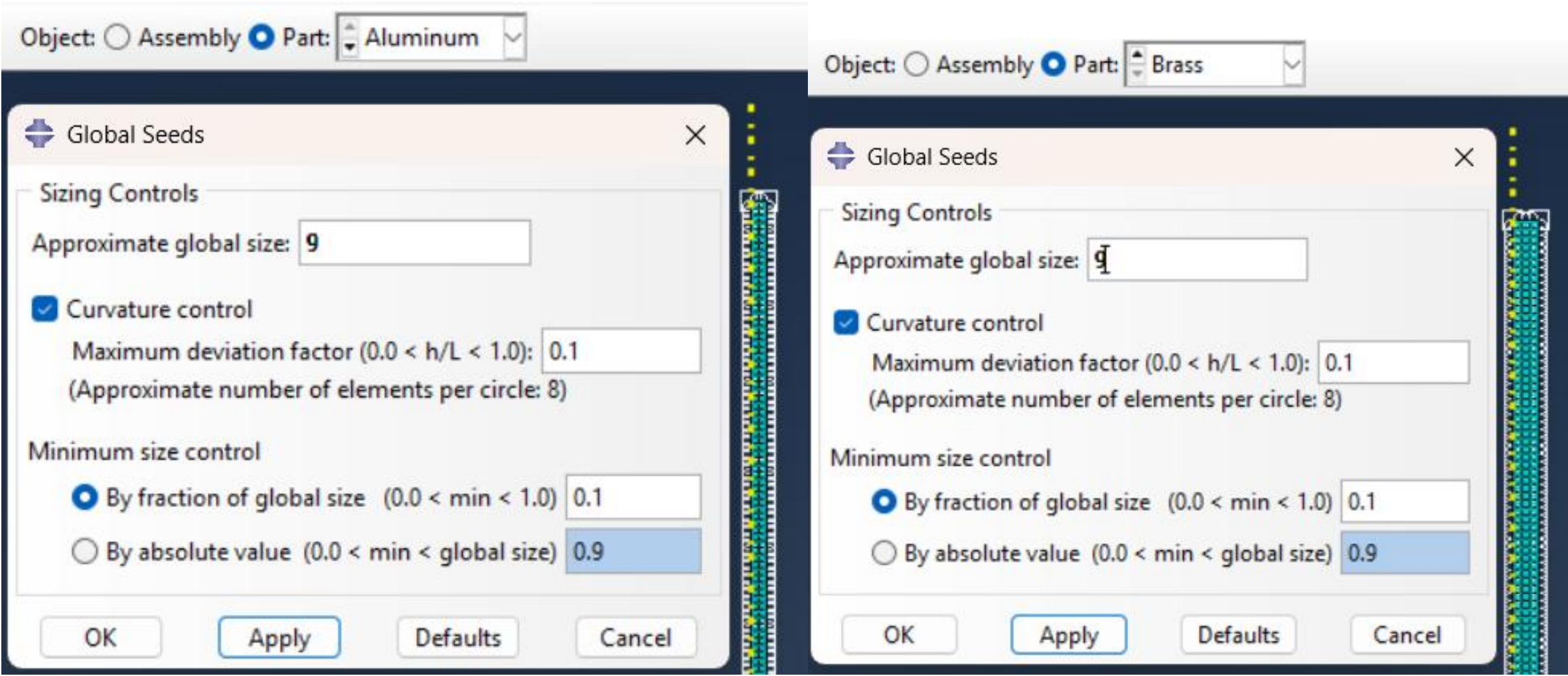
- Interaction



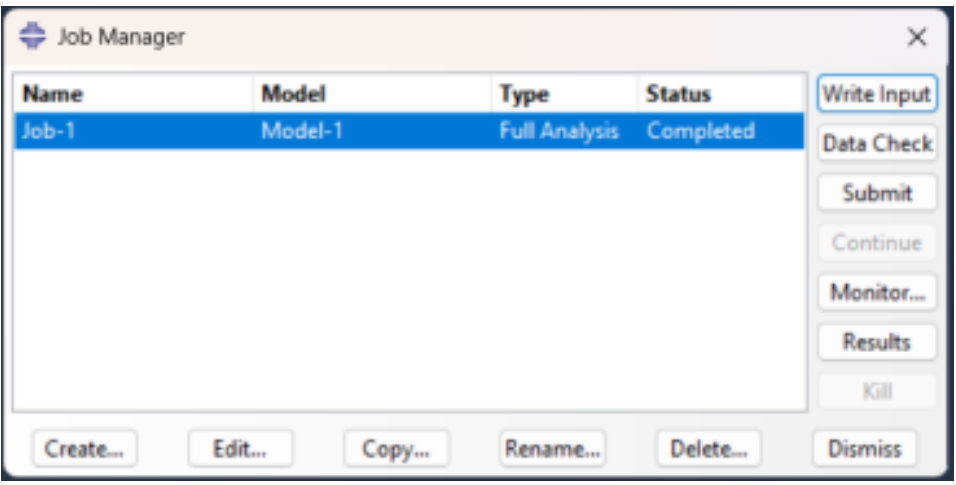
- Load



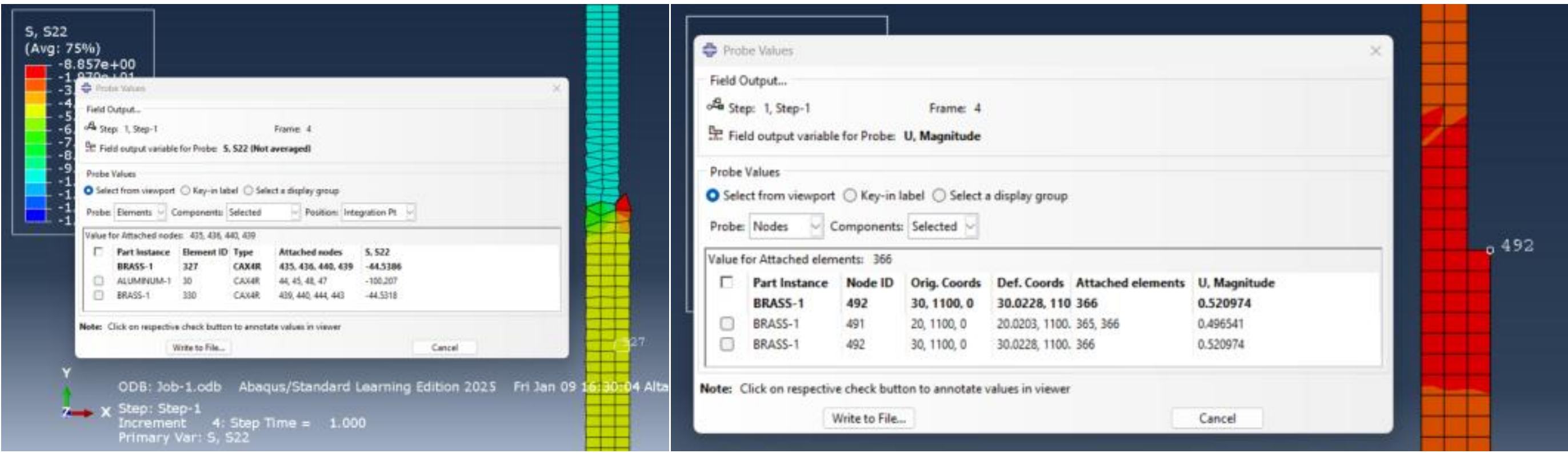
- Mesh



- Job



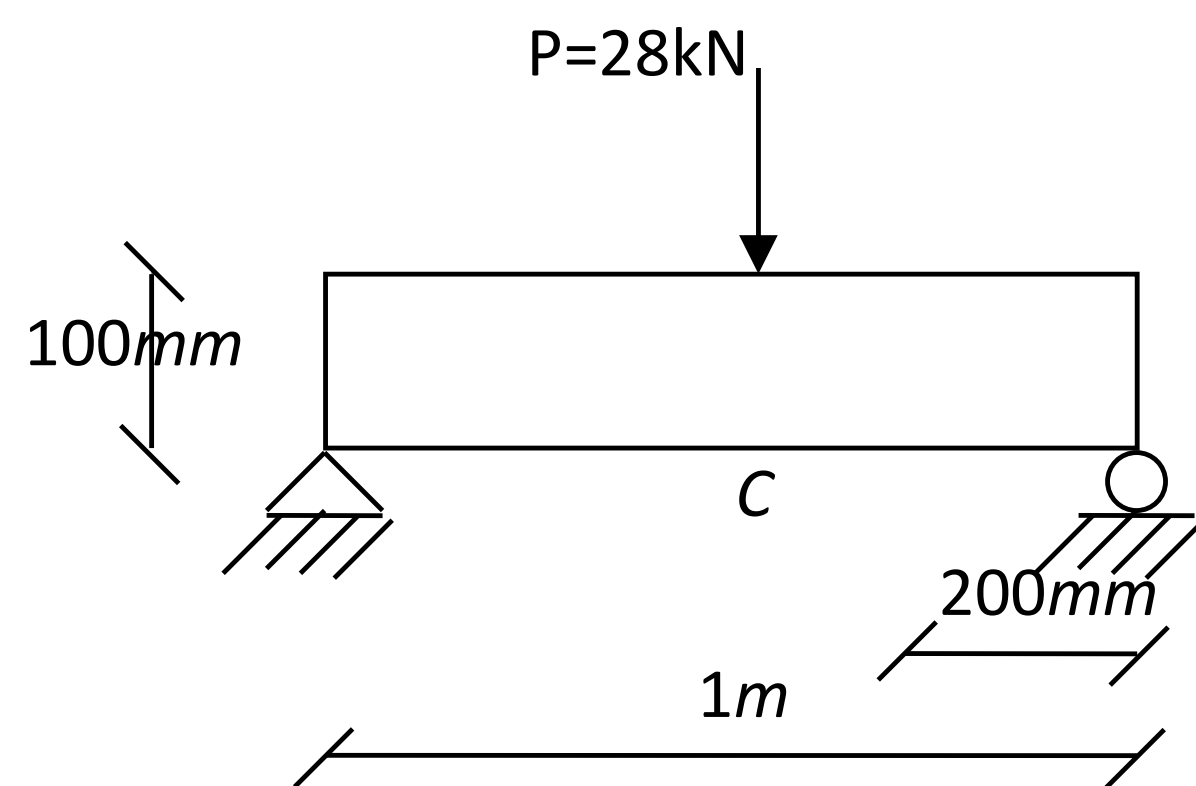
- Visualization and results



3. Compare the analytical and numerical solutions.

	1. Analytical solution:	2. Numerical solution:
Normal stress in AB section	$\sigma_x = -44.39$ MPa	$S_{11} = -44.53$ MPa
Normal stress in BC section	$\sigma_x = -99.74$ MPa	$S_{11} = -100.207$ MPa
Displacement at point B:	$\delta_B = 0.502$ mm	$U_1 = 0.52$ mm

B. Stress in beam



Width of the beam=25mm

Material Properties:

$E = 200 \text{ GPa}$, $\nu = 0.3$

$\sigma_{\text{allowable}} = 200 \text{ MPa}$

1. Analytical solution:

- Determine σ_x and τ_{xy} at point C. (C is in the middle and bottom of the beam)
- Determine the principal stresses σ_1 and σ_2 by drawing the state of stress at that point and using the formula or Mohr circle.
- Calculate the Von Mises stress at point C.

$$\sigma_v = \sqrt{\sigma_1^2 + \sigma_2^2 - \sigma_1 \sigma_2}$$

- What is the damage at point C? (Compare Von Mises stress with $\sigma_{\text{allowable}}$)

2. Numerical solution:

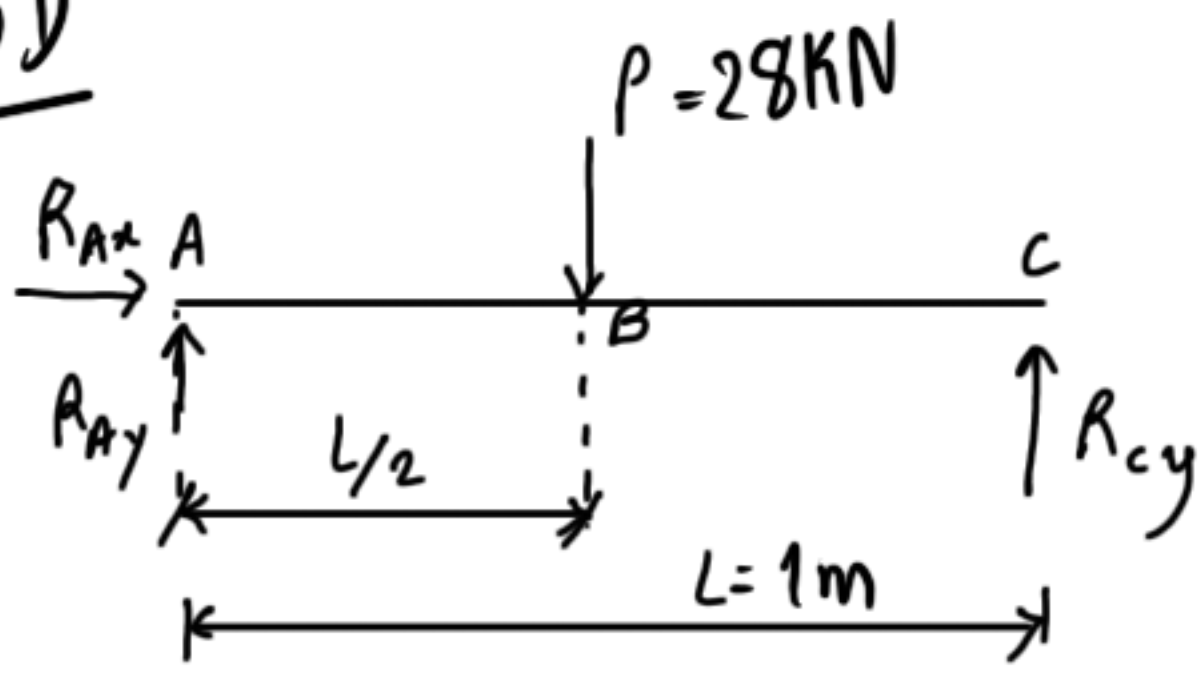
Using Abaqus Education Software, make a 2D planar model of the beam above. Determine the parameters below

	3. Analytical solution:	4. Numerical solution:
Normal stress at point C:	$\sigma_x = \dots\dots\dots \text{MPa}$	$S_{11} = \dots\dots\dots \text{MPa}$
Shear stress at point C:	$\tau_{xy} = \dots\dots\dots \text{MPa}$	$S_{12} = \dots\dots\dots \text{MPa}$
Principal stresses at point C: σ_x	$\sigma_1 = \dots\dots\dots \text{MPa}$ (Max) $\sigma_2 = \dots\dots\dots \text{MPa}$ (Min)	Max. Principal = MPa Min. Principal = MPa
Von Mises stress at point C	$\sigma_v = \dots\dots\dots \text{MPa}$	$S_{\text{Mises}} = \dots\dots\dots \text{MPa}$

3. Compare all parameters by both methods. If the results are not the same, please change your mesh size to as small as possible. Give your final comment.

1. Analytical Solution

FBD



Global Sys

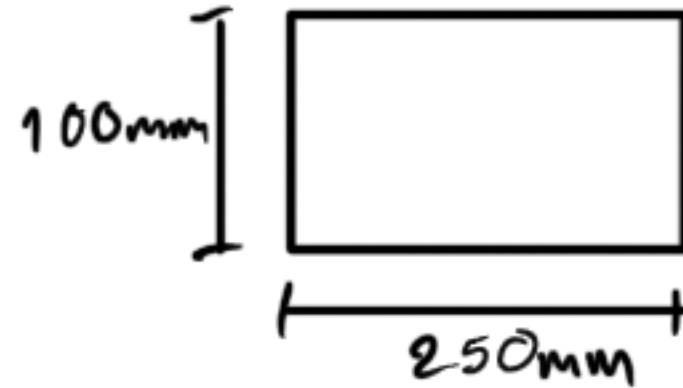


- Reaction Force

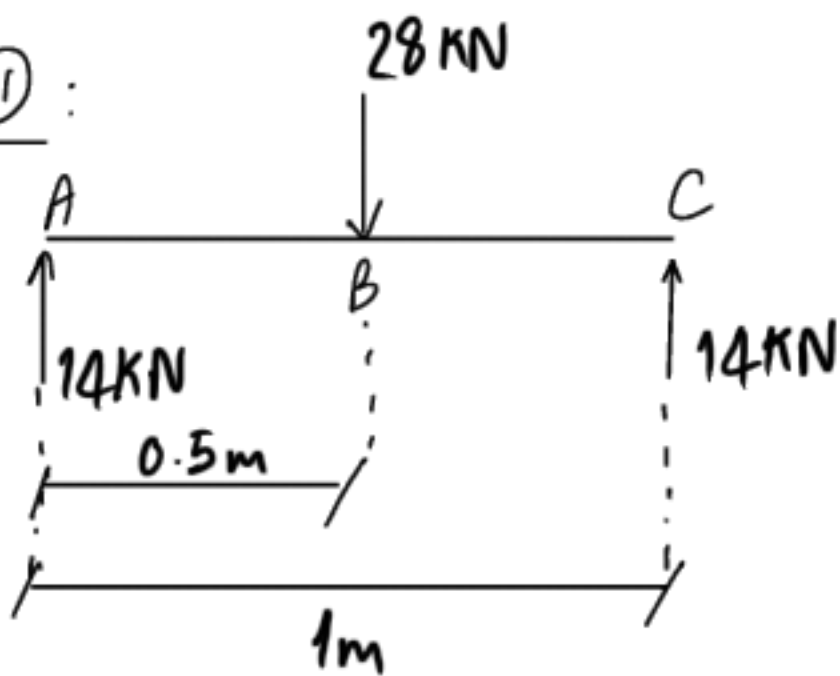
$$\begin{cases} \sum F_x = 0 \Rightarrow R_{Ax} = 0 \\ \sum F_y = 0 \Rightarrow R_{Ay} + R_{Cy} = P \\ \sum M_O = 0 \Rightarrow R_{Cy} \times L - P \times \frac{L}{2} = 0 \end{cases}$$

$$\Rightarrow R_{Cy} = \frac{P \times \frac{L}{2}}{L} = \frac{P}{2}$$

$$\Rightarrow R_{Ay} = \frac{P}{2}$$

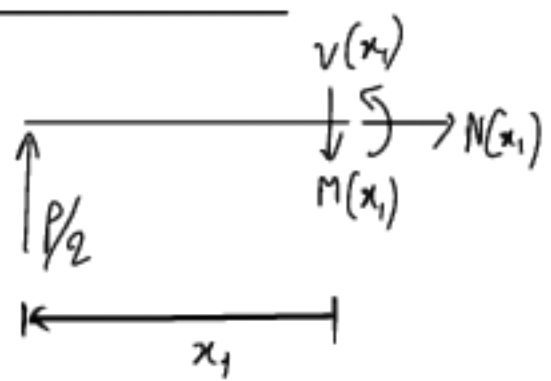


⇒ FBD:



+ Zonal Forces: 2 zones

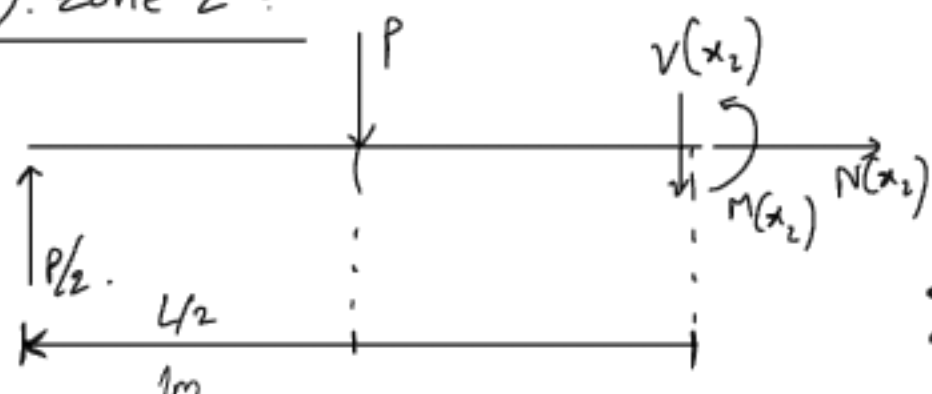
⊕ Zone 1:



$$\begin{aligned} \sum F_x = 0 &\Rightarrow N(x_1) = 0 \\ \sum F_y = 0 &\Rightarrow V(x_1) = \frac{P}{2} \\ \sum M = 0 &\Rightarrow M(x_1) - \frac{P}{2} \times x_1 = 0 \\ &\Rightarrow M(x_1) = \frac{P}{2} \times x_1 \end{aligned}$$

$$x \in]0, \frac{1}{2}[$$

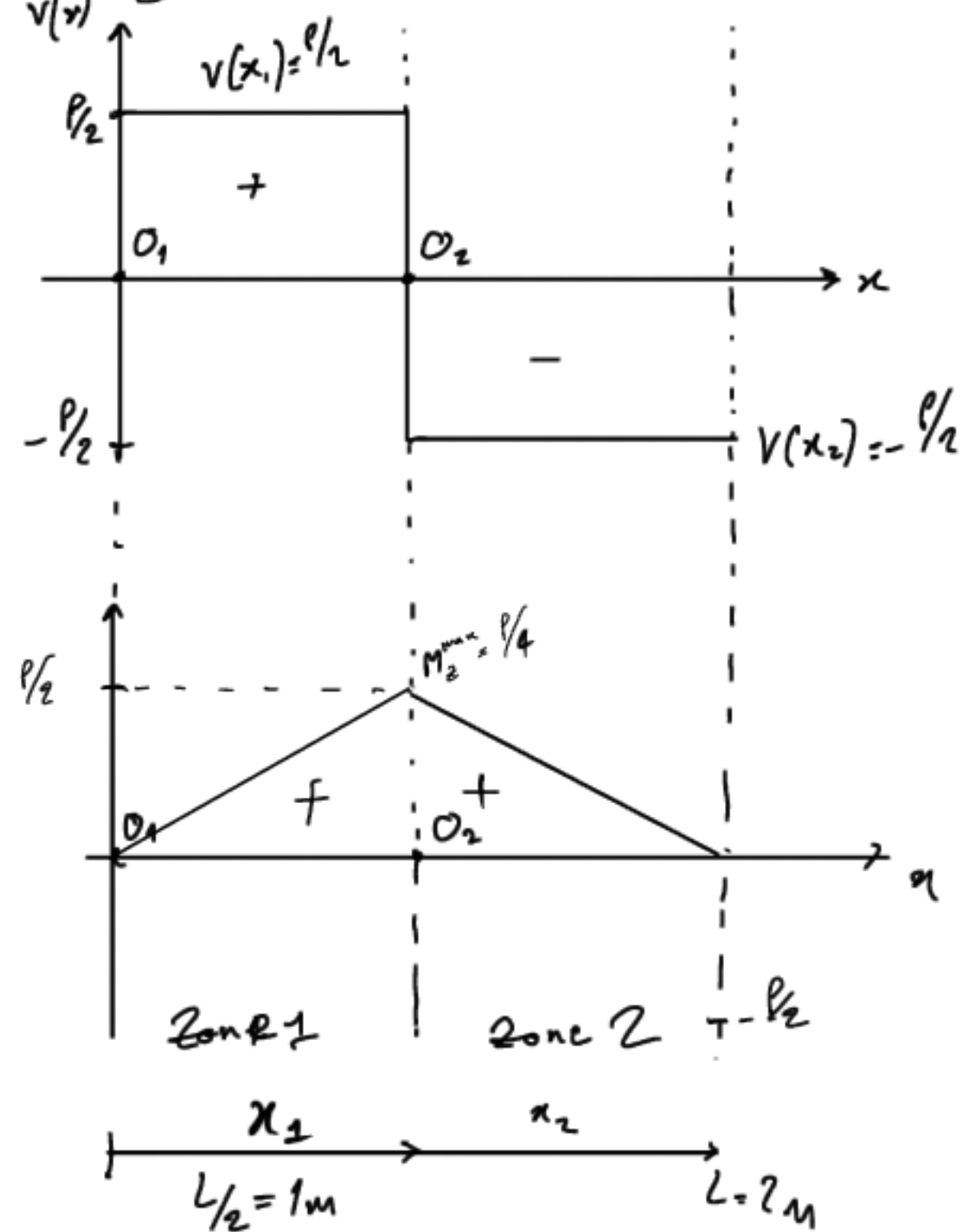
⊕ Zone 2:



$$\begin{aligned} \sum F_x = 0 &\Rightarrow N(x_2) = 0 \\ \sum F_y = 0 &\Rightarrow -V(x_2) - P + \frac{P}{2} = 0 \\ &\Rightarrow V(x_2) = -\frac{P}{2} \\ \sum M = 0 &\Rightarrow M(x_2) + P(x_2 - \frac{L}{2}) - \frac{P}{2}x_2 = 0 \\ &\Rightarrow M(x_2) = \frac{P x_2}{2} - P x_2 + \frac{P}{2} \\ &\Rightarrow M(x_2) = -\frac{P x_2}{2} + \frac{P}{2} \end{aligned}$$

$$x \in]\frac{1}{2}, L[$$

+ Diagram V(x) and M(x)



x_1	0	0.5
$V(x_1) = \frac{P}{2}$	0	$\frac{P}{4}$

x_2	0.5	1
$M(x_2) = -\frac{P x_2}{2} + \frac{P}{2}$	$\frac{P}{4}$	0

a) Determine σ_x and τ_{xy} at point C.

$$\sigma_x = -\frac{M_z^c(x)}{I_{cz}} y^c$$

C is at $x = \frac{L}{2} = 0.5m$

$$\Rightarrow M_z^c(x=0.5) = \frac{P}{2} \times \frac{L}{2} = \frac{28}{4} = 7kN \cdot m = 7000N \cdot m$$

$$y^c = \frac{100}{2}mm = 0.05m$$

$$I_{cz} = \frac{bh^3}{12} = \frac{0.25 \times (0.1)^3}{12} = 2.083 \times 10^{-6}m^4$$

$$\sigma_x = \frac{-7000N \times 0.05}{2.083 \times 10^{-6}m^4} = +168.02MPa$$

$$\tau_{xy} = \frac{V_y^c(x=\frac{1}{2})}{I_{cz} \cdot b} Q(y=\frac{h}{2})$$

$$\text{Since } Q_y \int_{\Omega} y dA = \bar{y}_{\Omega} \times \Omega$$

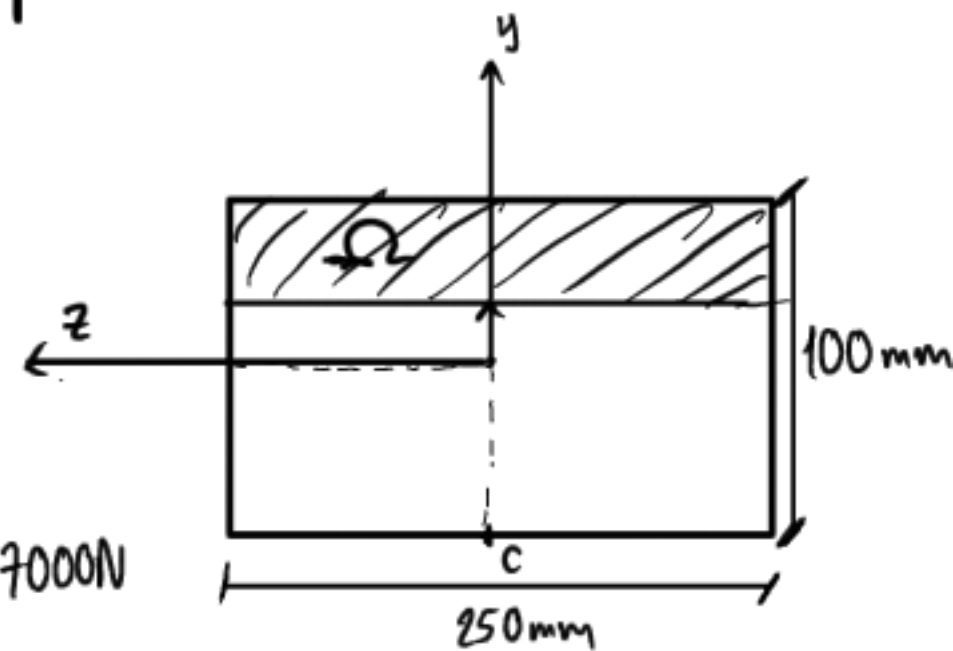
$$= \frac{\frac{h}{2} + y}{2} \times b \left(\frac{h}{2} - y \right)$$

$$= \frac{b}{2} \left[\left(\frac{h}{2} \right)^2 - y^2 \right]$$

$$\text{Since } Q(y=\frac{h}{2}) = \frac{b}{2} \left[\left(\frac{h}{2} \right)^2 - \left(\frac{h}{2} \right)^2 \right]$$

$$= 0$$

$$\Rightarrow \tau_{xy} = 0$$



$$\text{Since } \bar{y}_{\Omega} = \frac{y + \frac{h}{2}}{2}$$

$$\Omega = b \times \left(\frac{h}{2} - y \right)$$

b). Determine the principle stress σ_1 and σ_2

Using the formula from Mohr's circle:

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Since there is no load/external force in y direction at C

$$\Rightarrow \tau_y = 0$$

$$\sigma_{1,2} = 84.01 \pm \sqrt{\left(\frac{168-0}{2}\right)^2 + (0)^2} = 84.01 \pm 84.01$$

Thus

$\sigma_1 = 168.02 \text{ MPa}$	(maximum).
$\sigma_2 = 0 \text{ MPa}$	(minimum).

c). Calculate the Von Mises at point C

$$\begin{aligned}\sigma_v &= \sqrt{\sigma_1^2 + \sigma_2^2 - \sigma_1 \sigma_2} \\ &= \sqrt{0^2 + 168.02^2 - (168.02)(0)} \\ \sigma_v &= 168.02 \text{ MPa}\end{aligned}$$

d). What is the damage at point C?

+ We know $\sigma_{\text{allowable}} = 200 \text{ MPa}$

$$\sigma_v = 168.02 \text{ MPa}$$

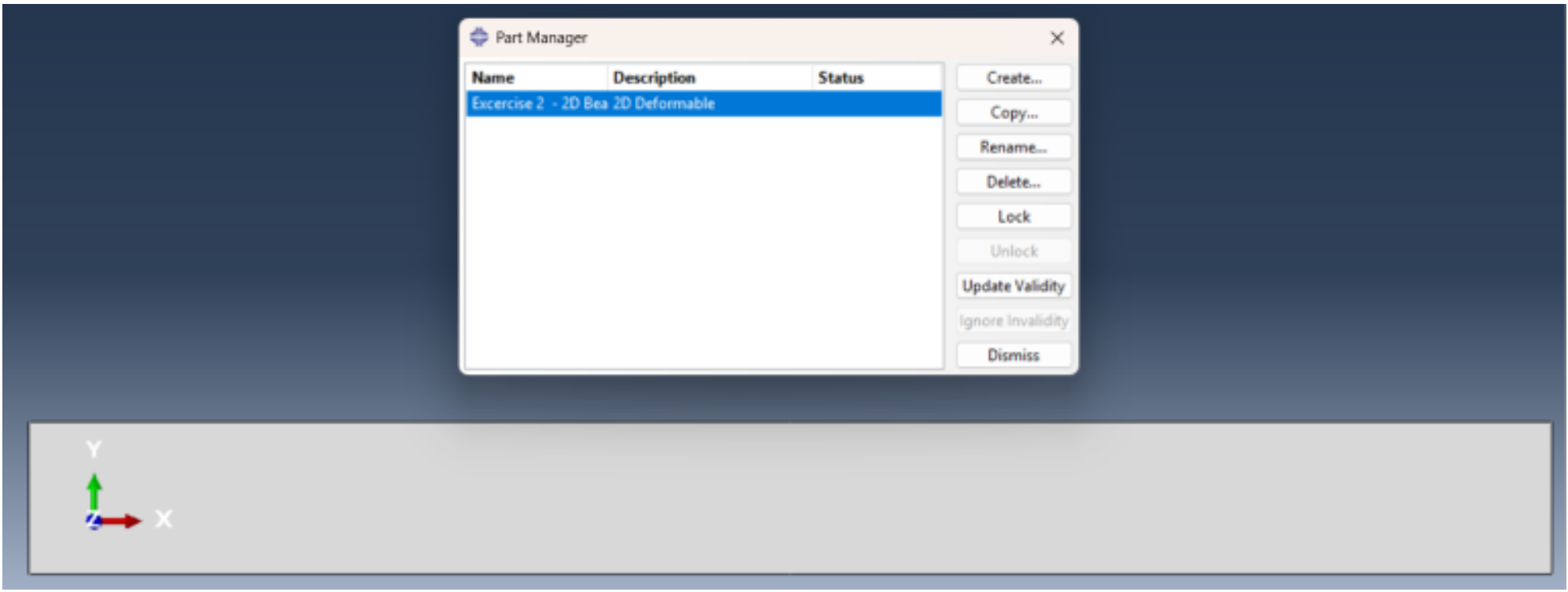
Since $\sigma_v < \sigma_{\text{allowable}}$

Thus, the material will not fail under load at point C.

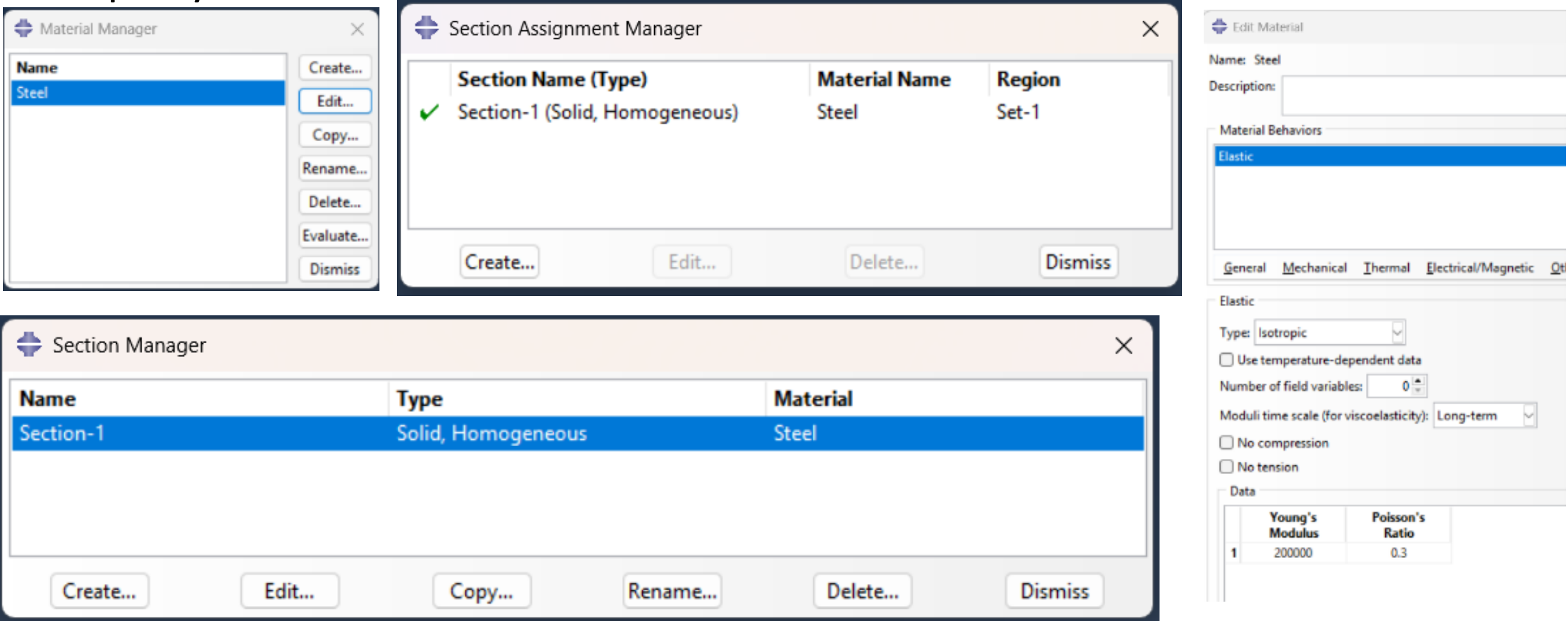
2. Numerical solution:

Using Abaqus Education Software, make a 2D planar model of the beam above. Determine the parameters below

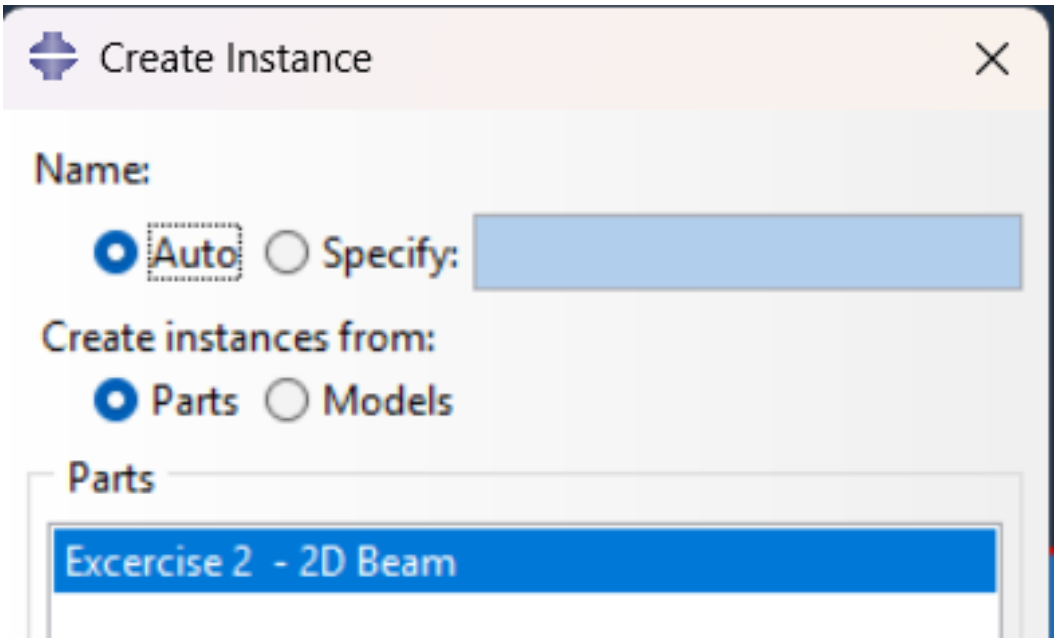
- First define the parts:



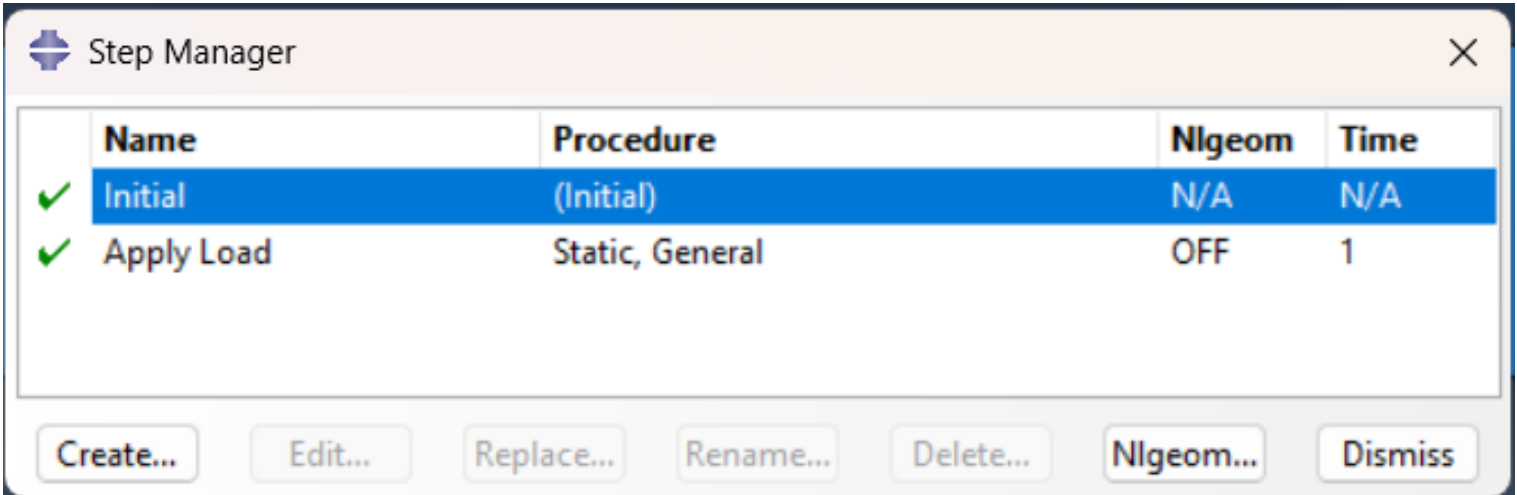
- Property



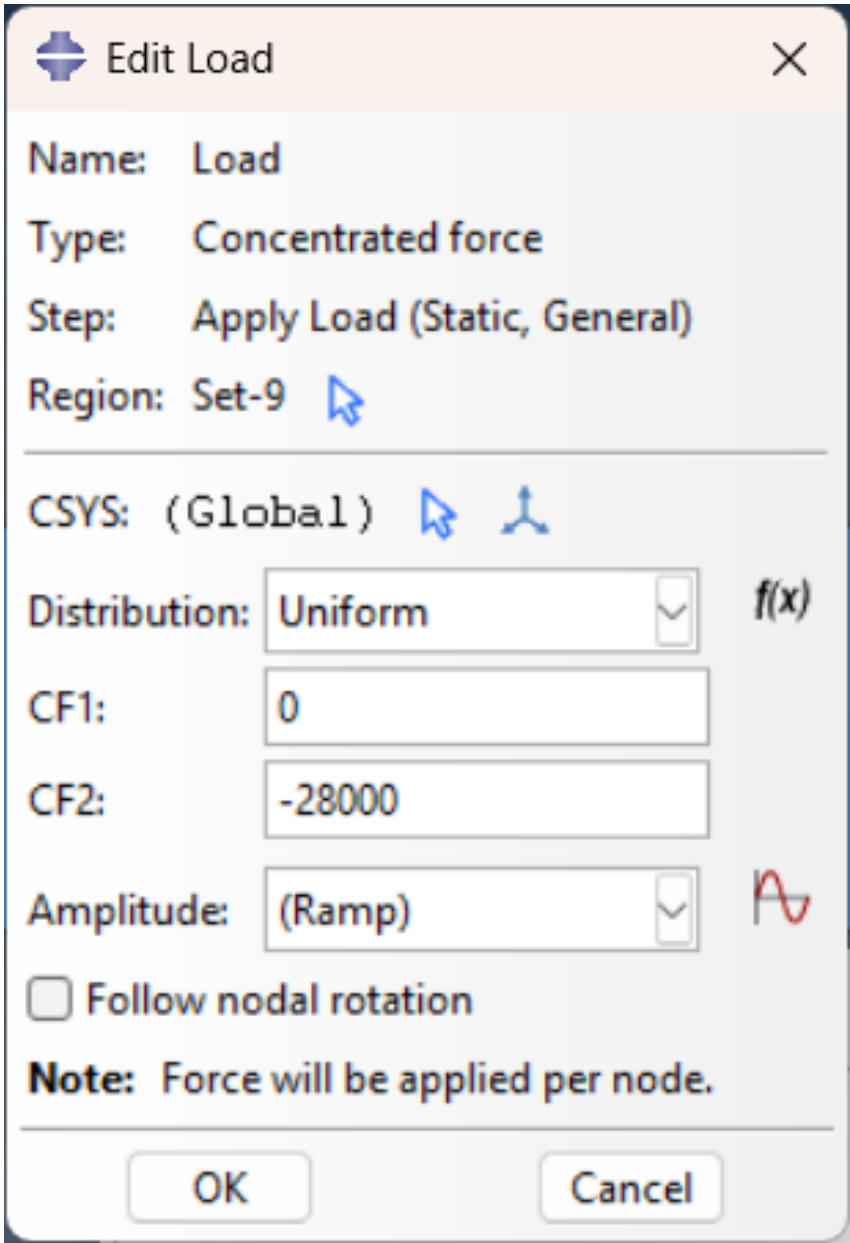
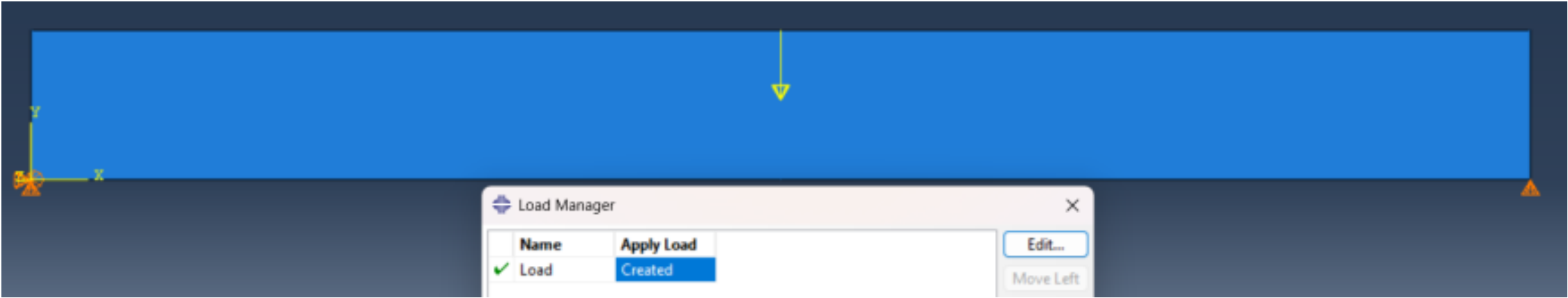
- Assemble



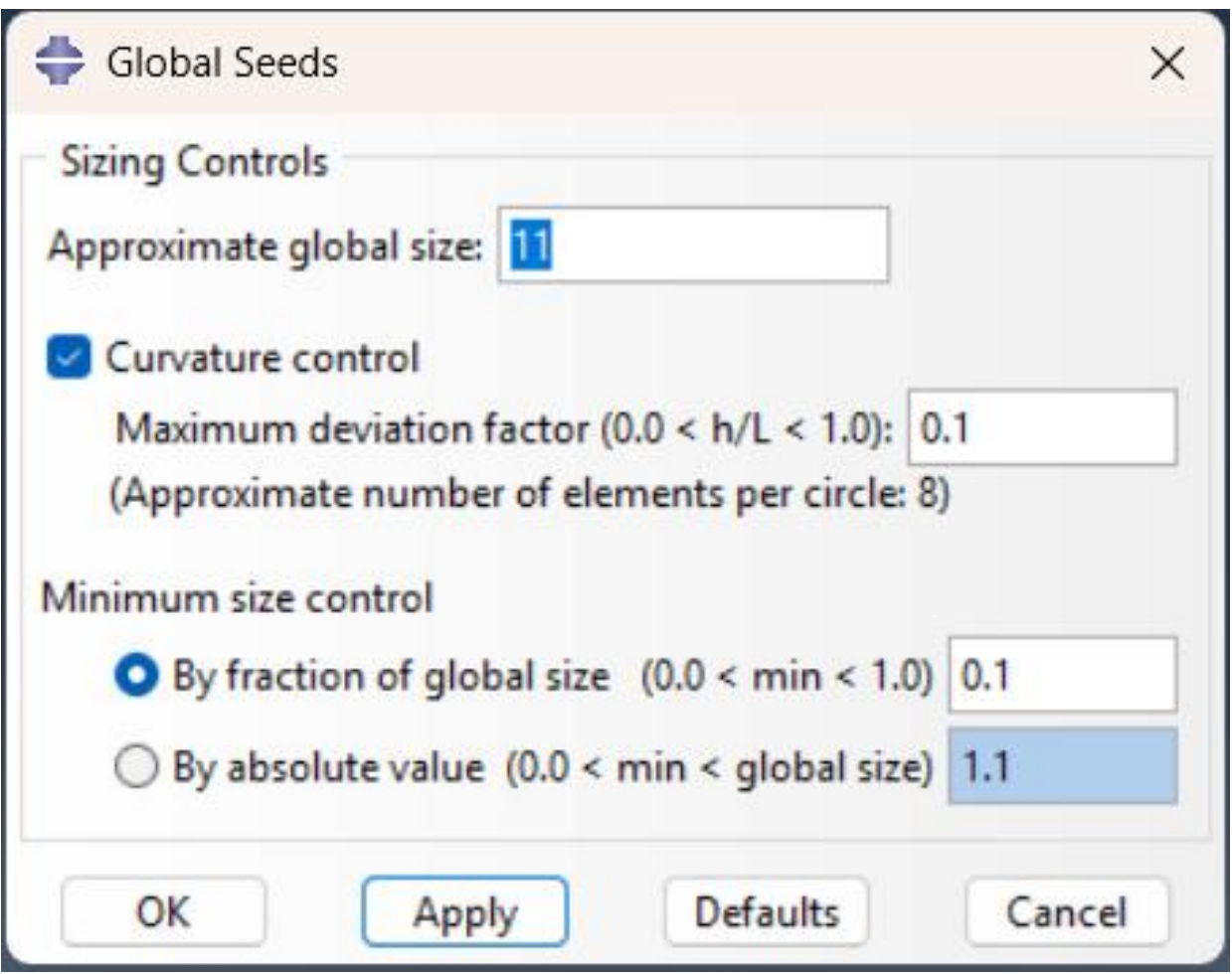
- Step



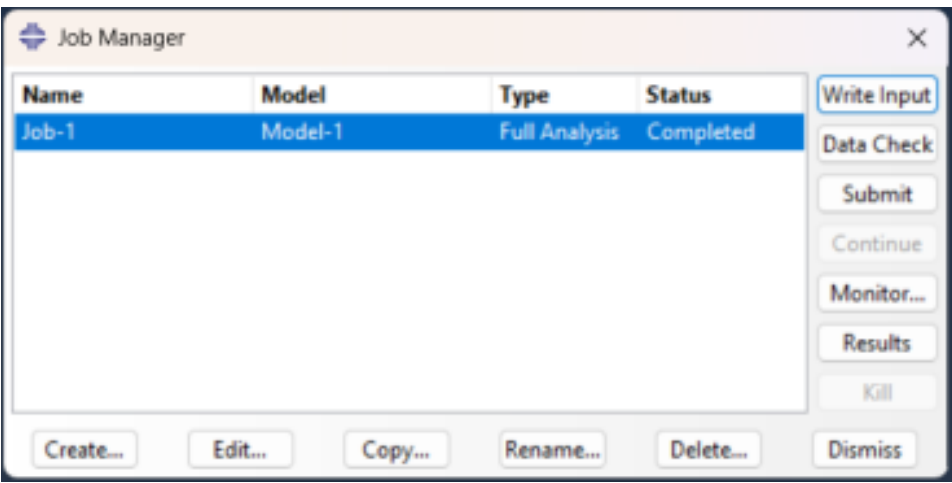
- Load



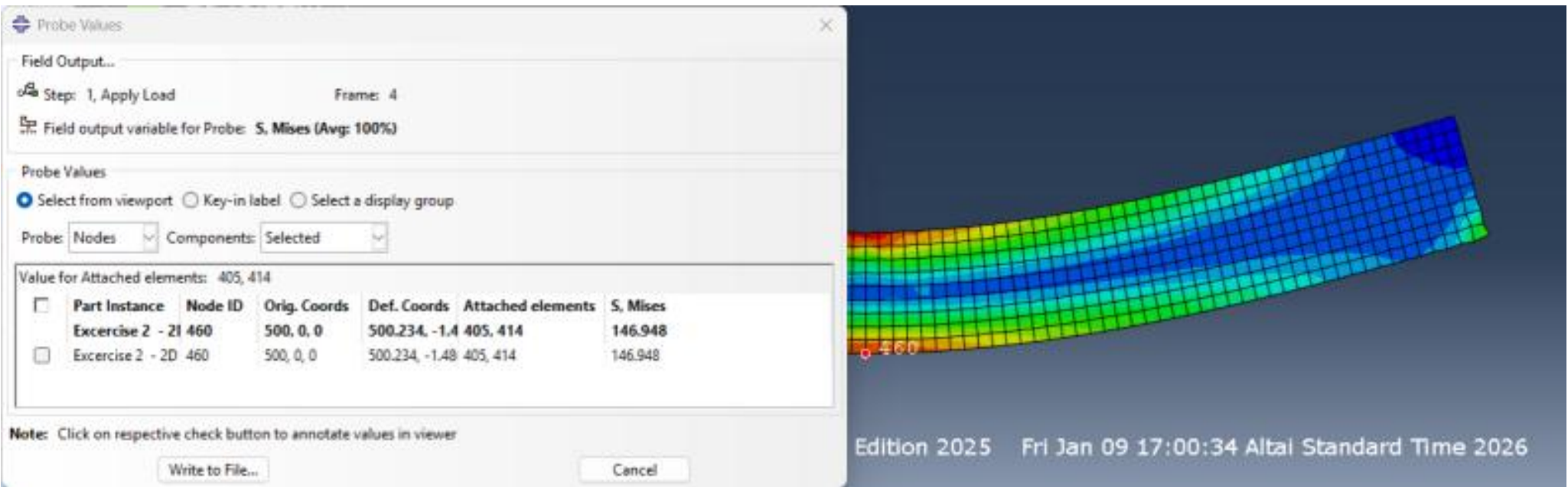
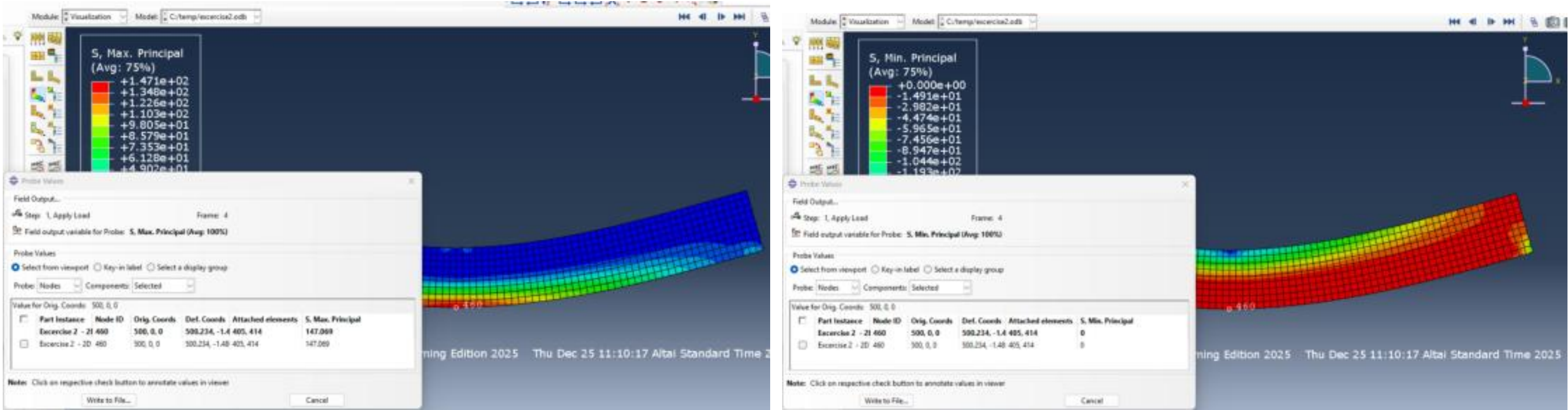
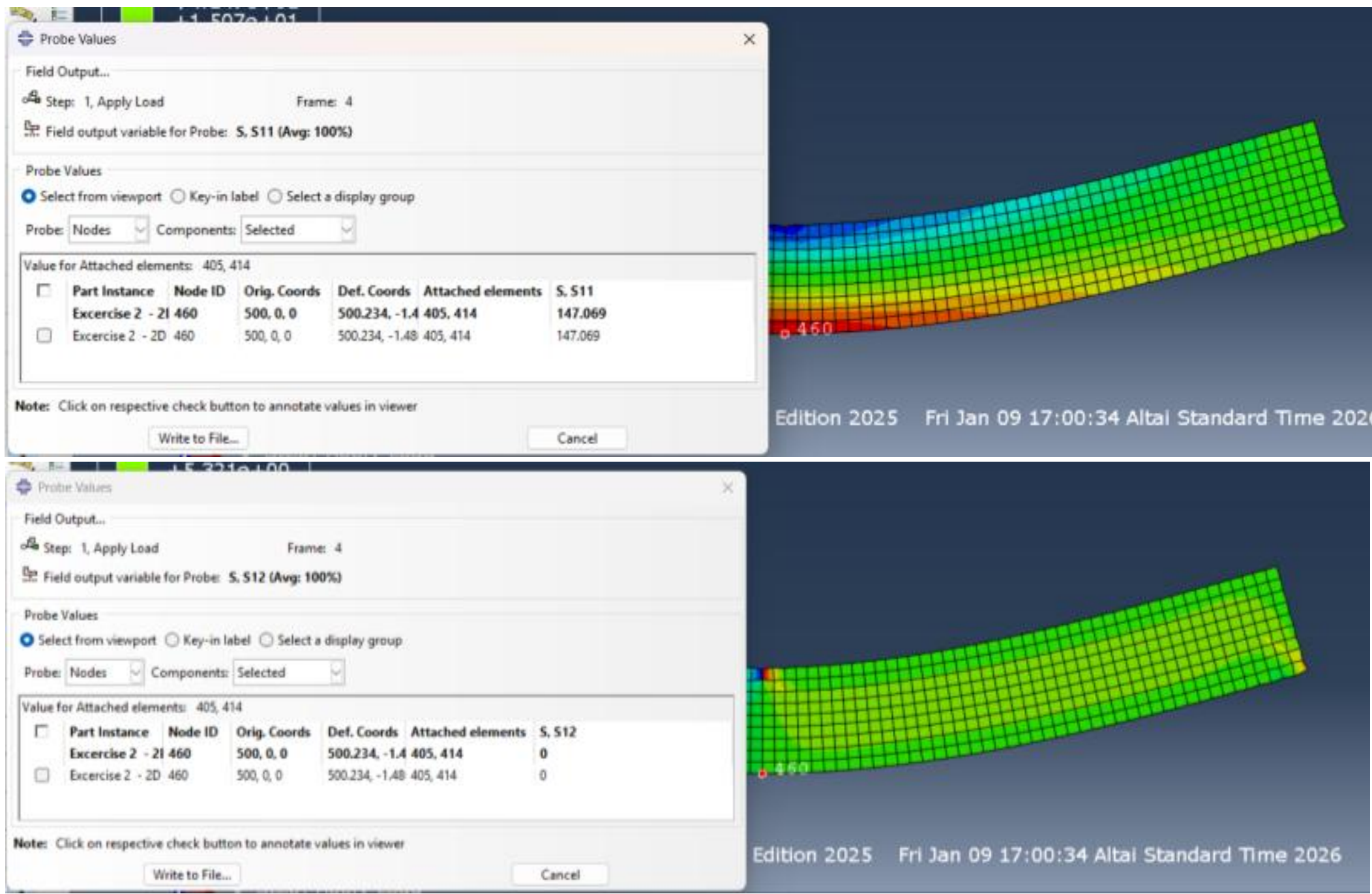
- Mesh



- Job



- Visualization and results



	3. Analytical solution:	4. Numerical solution:
Normal stress at point C:	$\sigma_x = 16.8$ MPa	$S_{11} = 147.07$ MPa
Shear stress at point C:	$\tau_{xy} = 0$ MPa	$S_{12} = 0$ MPa
Principal stresses at point C: σ_x	$\sigma_1 = 16.8$ MPa (Max) $\sigma_2 = 0$ MPa (Min)	Max. Principal = 147.05 MPa Min. Principal = 0 MPa
Von Mises stress at point C	$\sigma_v = 16.8$ MPa	$S_{Mises} = 146.948$ MPa

3. Compare all parameters by both methods. If the results are not the same, please change your mesh size to as small as possible. Give your final comment.

Can't change mesh size any further, results close enough.